

A Mobile App to search for the food additives by computer vision

WONG HO FAI

22205942



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01

Background

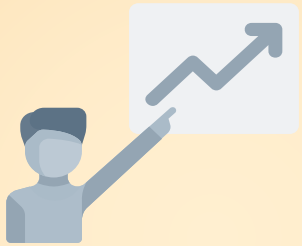




Project Background

- **Food additives**

✓ food safer
✓ tastier
✓ more appealing
✓ more nutritious
✓ extend its shelf life



But Also, may be harmful to specific people

Pregnant women



Children



People with specific food allergies







Project Background

- **Consumer**

✓ Hard to understand the additives code (E-number etc.) on food label.

✓ Lack of knowledge about the additives or ingredients.

✓ Complex searching step and time cost.

suffering in choosing the suitable food product when shopping



Problem

Current market applications







E code:

- **Only on food additives check**
- **No analysis and other function**

AI Calorie calculator:

- **No food additives check**
- **Only some side function**

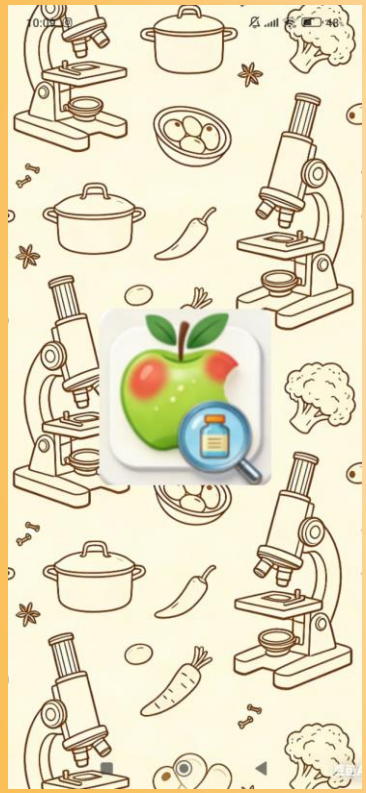
TagHalal:

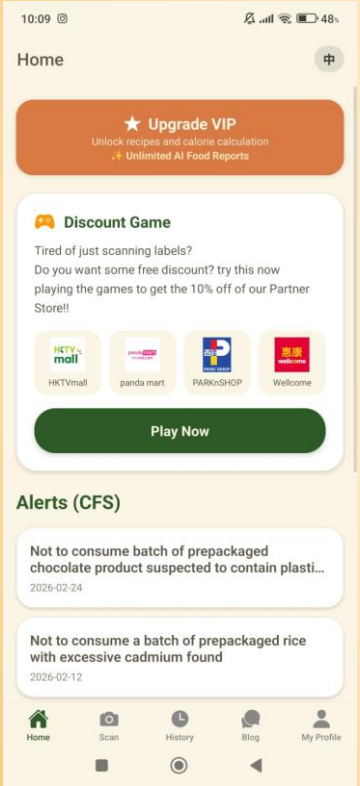
- **Only focus of Halal diet**

Function is split across different applications.



Solution





ScanEat

An all-in-one Mobile App to search for the food additives





Solution

1. **Core**
- **Food additives and ingredients scan and analysis**
2. **Additional related function**
- **Social media platforms (social value)**
- **Quiz game (Education & commercial value)**
- **Food safety alerts (social value)**
- **VIP feature (commercial value)**



Goal:

To have a all-in-one applications to letting the user can get rid of suffering on product selection when buying, also providing the value of social value, commercial value and education.



02

System Overview





System Implementation

- **Frontend:**

Expo + React Native

React Native Expo

- **Development tools:**

VS code

- **AI Model:**

Gemini

Gemini-3.1-pro (POE)

- **Backend:**

node.js

Node.js

- **Database:**

mongo DB

MongoDB



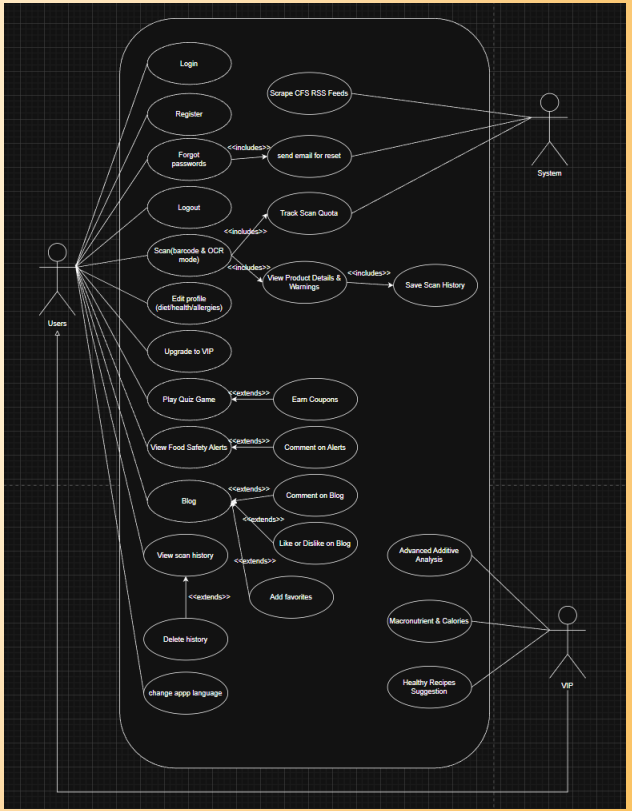
System Overview

User roles:

- **Normal User**
- **VIP User**



Use Case Diagram:



03

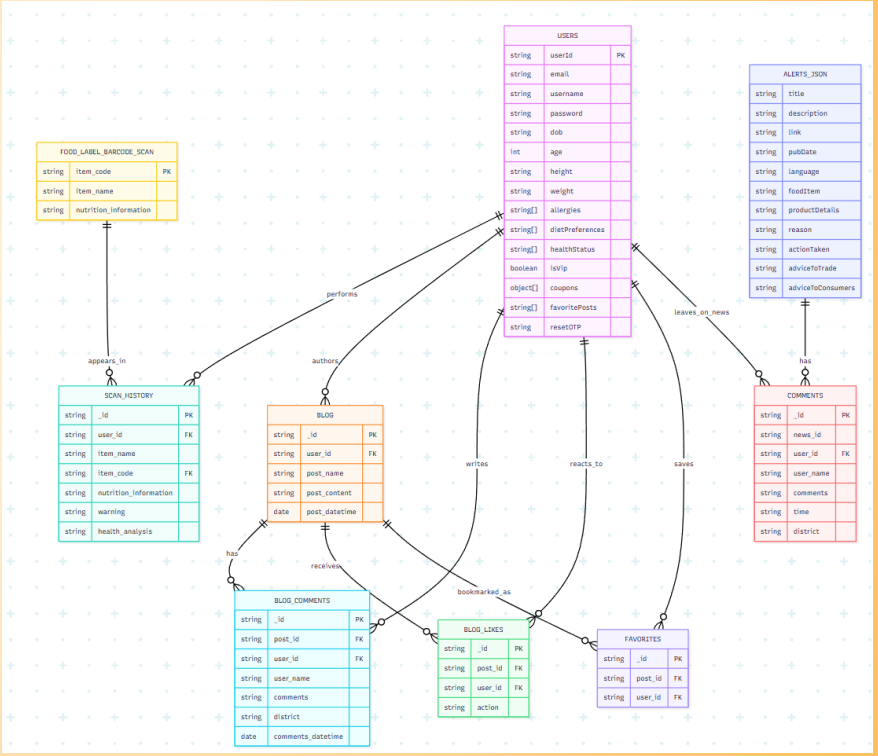
Database Design



Database Design

Total in 9 TABLEs:

1. **FoodLabelBarCodeScan**
2. **ScanHistory**
3. **Users**
4. **Blog**
5. **Comments (alert comments)**
6. **Blog_comments**
7. **Blog_Likes**
8. **Favorites (user favorites list)**
9. **Alert.json (no in mongoDB)**

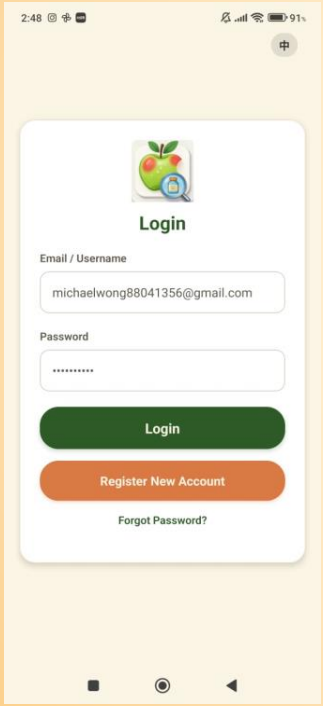


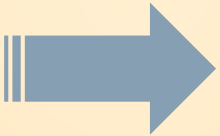
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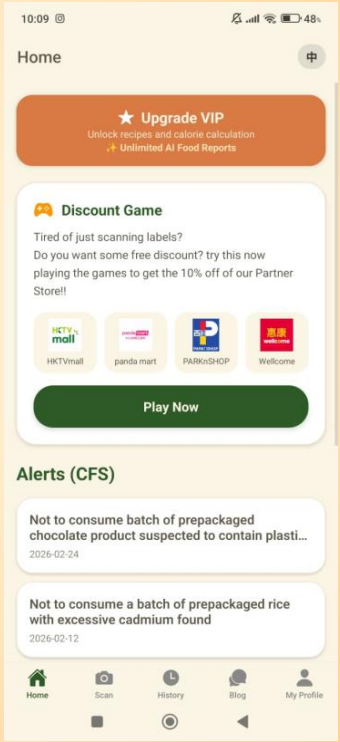
Storyboard



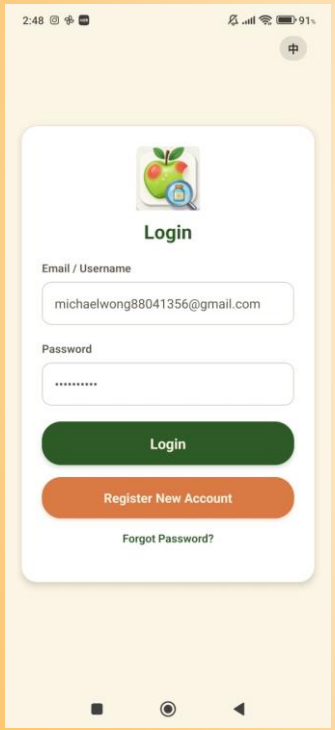
Login

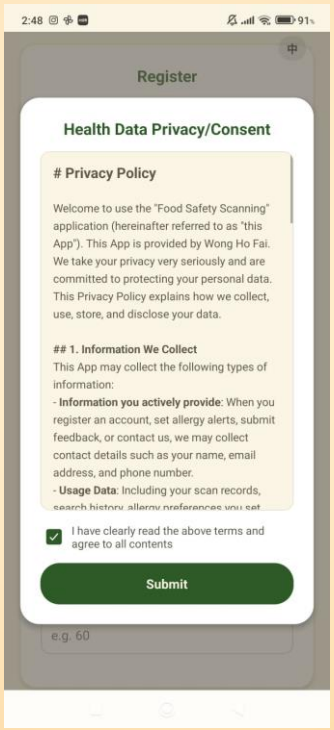

2:48
91%
中
Login
Email / Username
michaelwong88041356@gmail.com
Password
Login
Register New Account
Forgot Password?

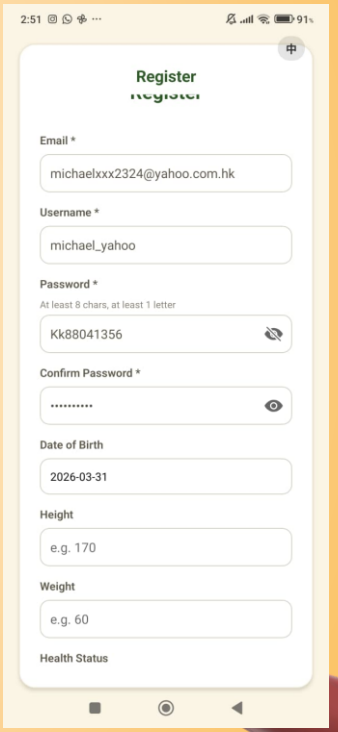



10:09
48%
Home
中
Upgrade VIP
Unlock recipes and calorie calculation
✨ Unlimited AI Food Reports
Discount Game
Tired of just scanning labels?
Do you want some free discount? try this now playing the games to get the 10% off of our Partner Store!!
HKTVmall
panda mart
PARKnSHOP
Wellcome
Play Now
Alerts (CFS)
Not to consume batch of prepackaged chocolate product suspected to contain plasti...
2026-02-24
Not to consume a batch of prepackaged rice with excessive cadmium found
2026-02-12
Home
Scan
History
Blog
My Profile

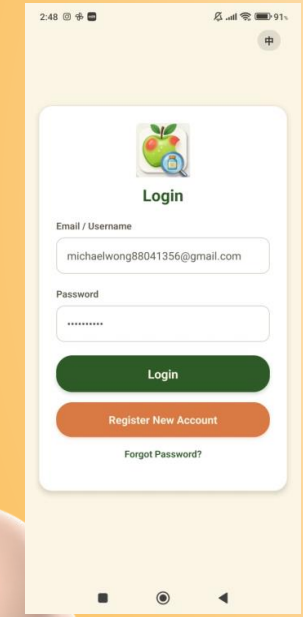
Register

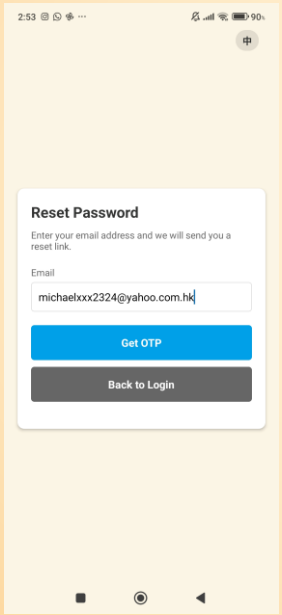

Login
Email / Username
michaelwong88041356@gmail.com
Password
Login
Register New Account
Forgot Password?

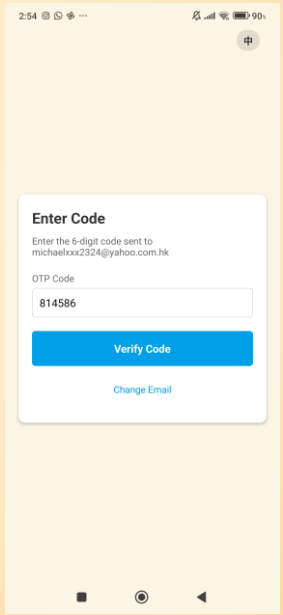

Register
Health Data Privacy/Consent
Privacy Policy
Welcome to use the "Food Safety Scanning" application (hereinafter referred to as "this App"). This App is provided by Wong Ho Fai. We take your privacy very seriously and are committed to protecting your personal data. This Privacy Policy explains how we collect, use, store, and disclose your data.
1. Information We Collect
This App may collect the following types of information:
- Information you actively provide: When you register an account, set allergy alerts, submit feedback, or contact us, we may collect contact details such as your name, email address, and phone number.
- Usage Data: Including your scan records,
I have clearly read the above terms and agree to all contents
Submit

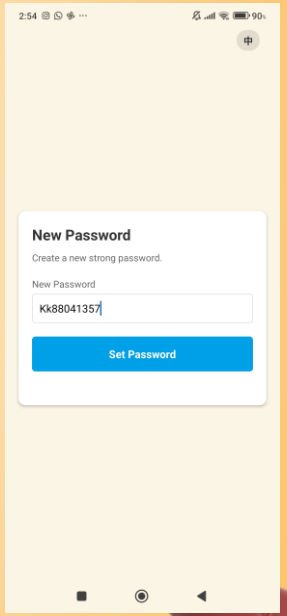

Register
Email *
michaelxxx2324@yahoo.com.hk
Username *
michael_yahoo
Password *
At least 8 chars, at least 1 letter
Kk88041356
Confirm Password *
Date of Birth
2026-03-31
Height
e.g. 170
Weight
e.g. 60
Health Status

Forgot Password





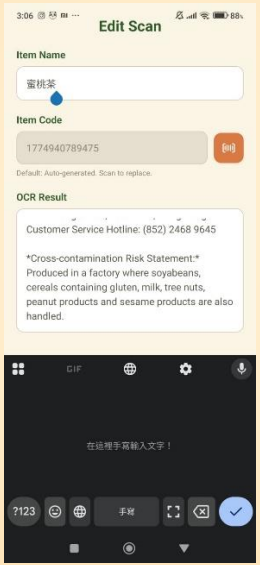


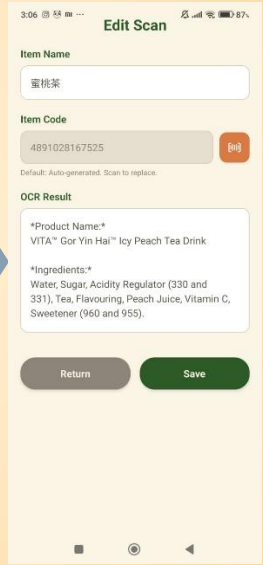


Scan


Scan
SCAN MODE
TEXT MODE
Remaining Quota: 3/5
Capture Text
Pick Image




Edit Scan
Item Name
蜜桃茶
Item Code
1774940789475
Default: Auto-generated. Scan to replace.
OCR Result
Customer Service Hotline: (852) 2468 9645
Cross-contamination Risk Statement:
Produced in a factory where soyabeans, cereals containing gluten, milk, tree nuts, peanut products and sesame products are also handled.

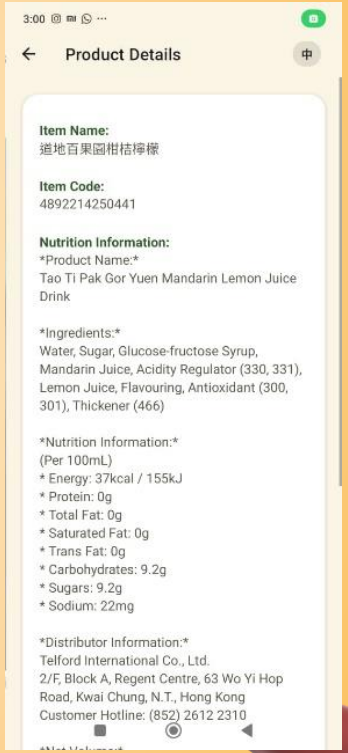

Edit Scan
Item Name
蜜桃茶
Item Code
4891028167525
Default: Auto-generated. Scan to replace.
OCR Result
Product Name:
VITA™ Gor Yin Hai™ Icy Peach Tea Drink
Ingredients:
Water, Sugar, Acidity Regulator (330 and 331), Tea, Flavouring, Peach Juice, Vitamin C, Sweetener (960 and 955).
Return
Save


Product Details
DANGER
DANGER: This product is manufactured in a facility that handles tree nuts and peanuts, which poses a severe risk to your nut allergy.
Item Name:
蜜桃茶
Item Code:
4891028167525
Nutrition Information:
Product Name:
VITA™ Gor Yin Hai™ Icy Peach Tea Drink
Ingredients:
Water, Sugar, Acidity Regulator (330 and 331), Tea, Flavouring, Peach Juice, Vitamin C, Sweetener (960 and 955).
Nutrition Information:
(Per 100mL)
- Energy: 20 kcal
- Protein: 0 g
- Fat, total: 0 g
- Saturated fat: 0 g
- Trans fat: 0 g
- Carbohydrates: 5.1 g
- Sugars: 4.8 g
- Sodium: 25 mg

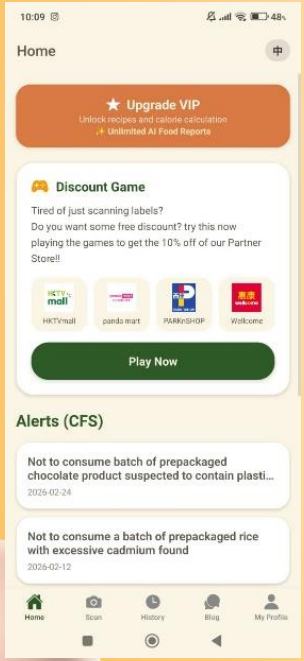
History

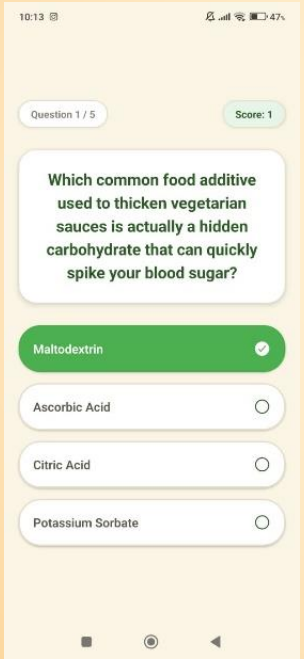

10:09
Home
中
Alerts (CFS)
Not to consume batch of prepackaged chocolate product suspected to contain plasti...
2026-02-24
Not to consume a batch of prepackaged rice with excessive cadmium found
2026-02-12
Not to consume batch of powdered infant formula
2026-02-03
Recent Scans
9314807025267
維他命糖
View
4892214250441
道地百果園柑桔檸檬
View
Home
Scan
History
Blog
My Profile

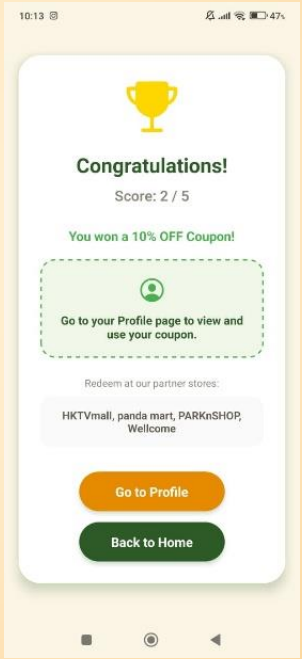

3:00
History
中
History Screen
Search by Name or Code
4892214250441
道地百果園柑桔檸檬
View
Home
Scan
History
Blog
My Profile

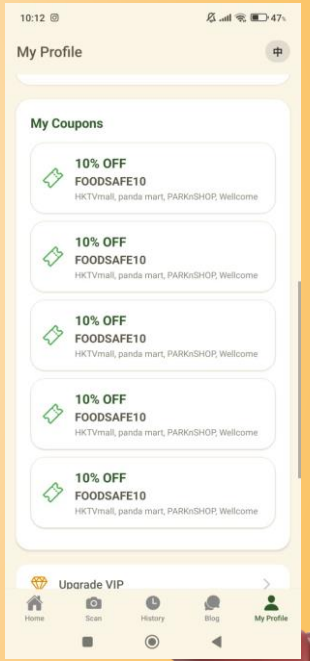

3:00
Product Details
中
Item Name:
道地百果園柑桔檸檬
Item Code:
4892214250441
Nutrition Information:
Product Name:
Tao Ti Pak Gor Yuen Mandarin Lemon Juice Drink
Ingredients:
Water, Sugar, Glucose-fructose Syrup, Mandarin Juice, Acidity Regulator (330, 331), Lemon Juice, Flavouring, Antioxidant (300, 301), Thickener (466)
Nutrition Information:
(Per 100mL)
* Energy: 37kcal / 155kJ
* Protein: 0g
* Total Fat: 0g
* Saturated Fat: 0g
* Trans Fat: 0g
* Carbohydrates: 9.2g
* Sugars: 9.2g
* Sodium: 22mg
Distributor Information:
Telford International Co., Ltd.
2/F, Block A, Regent Centre, 63 Wo Yi Hop Road, Kwai Chung, N.T., Hong Kong
Customer Hotline: (852) 2612 2310

Discount Game





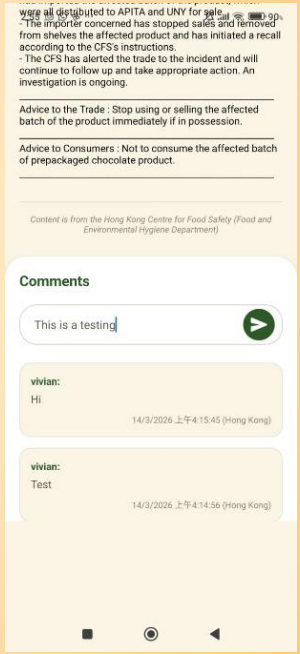




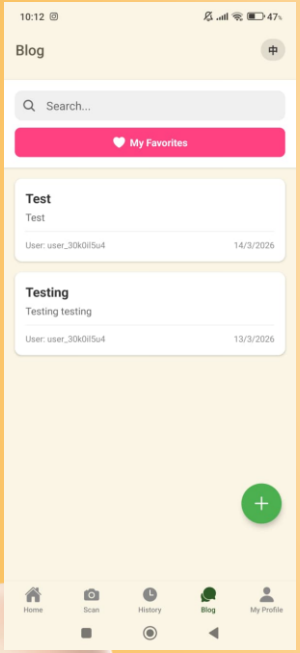
Alert News

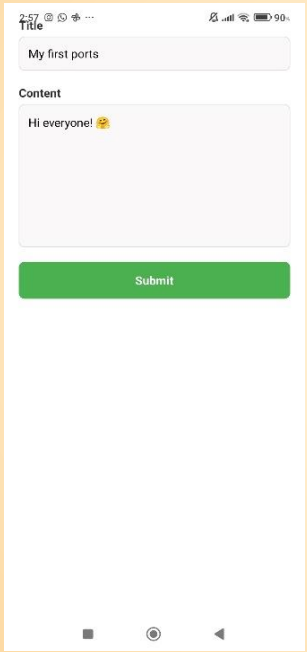

Home
Tired of just scanning labels?
Do you want some free discount? try this now playing the games to get the 10% off of our Partner Store!!
HKTVmall
panda mart
PARKnSHOP
Wellcome
Play Now
Alerts (CFS)
Not to consume batch of prepackaged chocolate product suspected to contain plasti...
2026-02-24
Not to consume a batch of prepackaged rice with excessive cadmium found
2026-02-12
Not to consume batch of powdered infant formula
2026-02-03
Home
Scan
History
Blog
My Profile

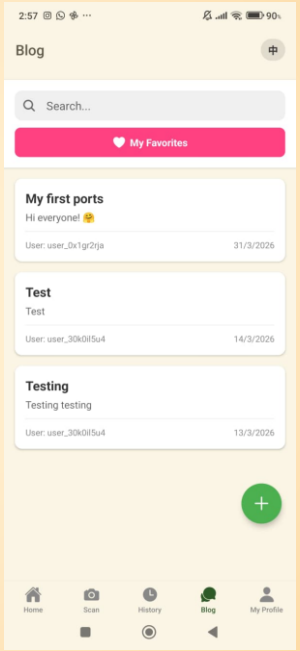

Return
Not to consume batch of prepackaged chocolate product suspected to contain plastic fragments
2026-02-24
Issue Date : 24.2.2026
Source : The Centre for Food Safety (CFS)
Food Product : Prepackaged chocolate product
Product Name and Description : Product name: Furuta Doremi Song Chocolate
Brand: Furuta
Place of origin: Japan
Pack size: 122 grams per pack
Batch number: 2026.10/JG+FM
Importer: Man Tung Trading Company Limited
Retailer: APITA and UNY
Best-before date: End of October, 2026
Reason For Issuing Alert : The CFS, through its Food Incident Surveillance System, noted a notification from the Japanese authority that the above-mentioned product may contain plastic fragments and is being recalled.
Action Taken : - Upon learning of the incident, the CFS immediately contacted local major importers and retailers for follow-up.
- According to preliminary investigation, the above importer had imported the affected batch of the product, which were all distributed to APITA and UNY for sale.
- The importer concerned has stopped sales and removed from shelves the affected product and has initiated a recall according to the CFS's instructions.
- The CFS has alerted the trade to the incident and will continue to follow up and take appropriate action. An investigation is ongoing.
Advice to the Trade : Stop using or selling the affected

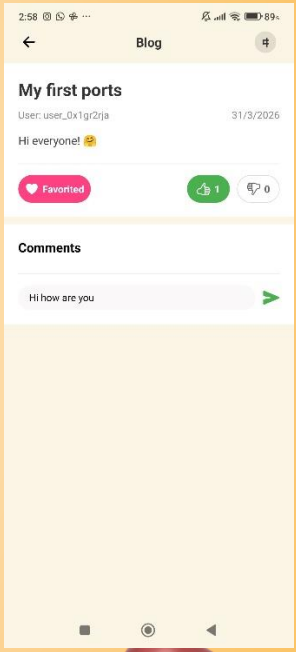

were all distributed to APITA and UNY for sale.
- The importer concerned has stopped sales and removed from shelves the affected product and has initiated a recall according to the CFS's instructions.
- The CFS has alerted the trade to the incident and will continue to follow up and take appropriate action. An investigation is ongoing.
Advice to the Trade : Stop using or selling the affected batch of the product immediately if in possession.
Advice to Consumers : Not to consume the affected batch of prepackaged chocolate product.
Content is from the Hong Kong Centre for Food Safety (Food and Environmental Hygiene Department)
Comments
This is a testing
vivian:
Hi
14/3/2026 上午4:15:45 (Hong Kong)
vivian:
Test
14/3/2026 上午4:14:56 (Hong Kong)

Blog

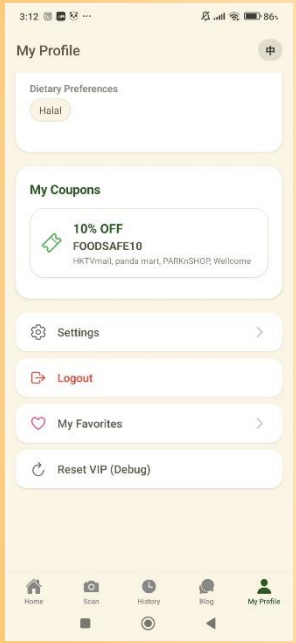

10:12
Blog
Search...
My Favorites
Test
Test
User: user_30k0il5u4
14/3/2026
Testing
Testing testing
User: user_30k0il5u4
13/3/2026
Home
Scan
History
Blog
My Profile


Title
My first ports
Content
Hi everyone! 🤗
Submit

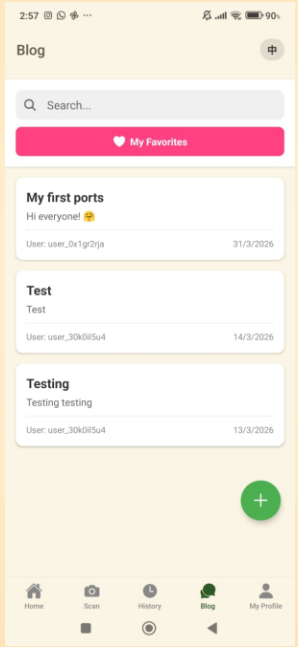

2:57
Blog
Search...
My Favorites
My first ports
Hi everyone! 🤗
User: user_0x1gr2rja
31/3/2026
Test
Test
User: user_30k0il5u4
14/3/2026
Testing
Testing testing
User: user_30k0il5u4
13/3/2026
Home
Scan
History
Blog
My Profile

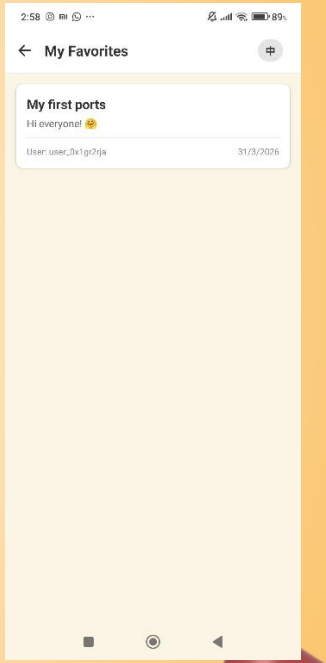

2:58
Blog
My first ports
User: user_0x1gr2rja
31/3/2026
Hi everyone! 🤗
Favorited
1
0
Comments
Hi how are you

Favourite

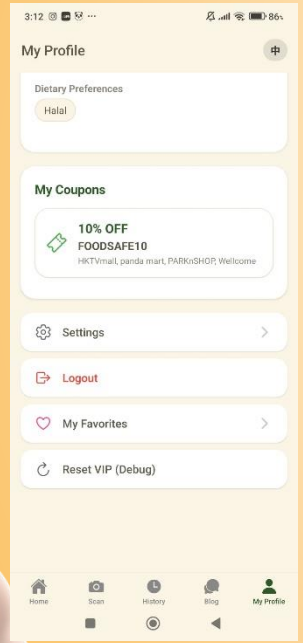


OR

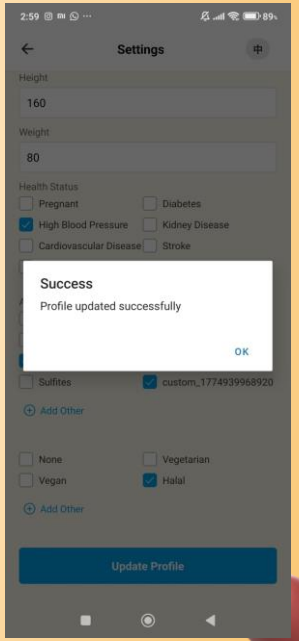




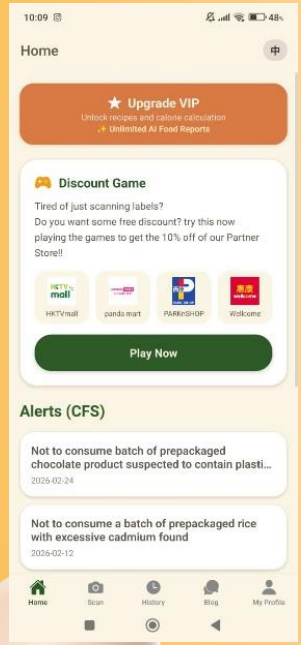
Profile Setting

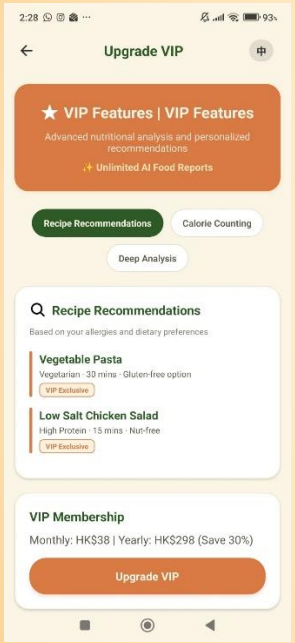


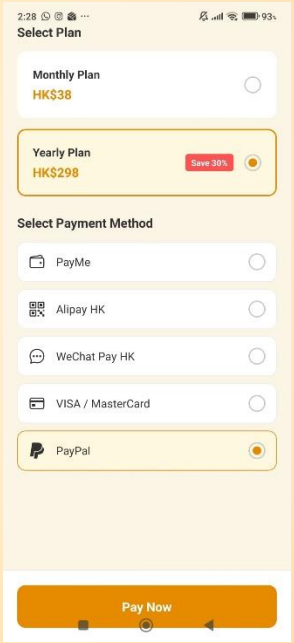


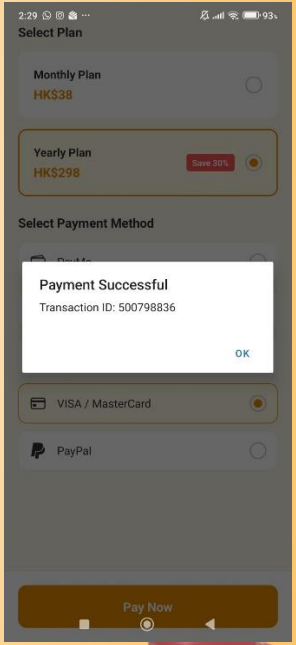


VIP Upgrade

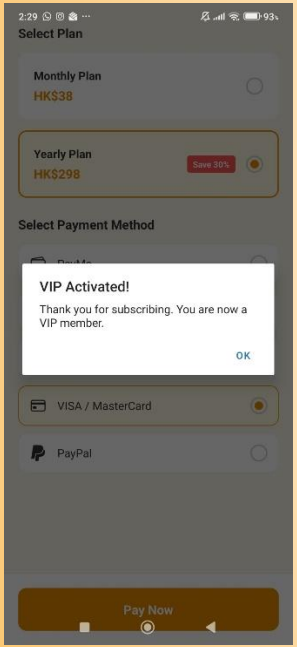

Home
Upgrade VIP
Unlock recipes and calorie calculation
✨ Unlimited AI Food Reports
Discount Game
Tired of just scanning labels?
Do you want some free discount? try this now playing the games to get the 10% off of our Partner Store!!
HKTVmall
PARKnSHOP
Play Now
Alerts (CFS)
Not to consume batch of prepackaged chocolate product suspected to contain plasti...
2026-02-24
Not to consume a batch of prepackaged rice with excessive cadmium found
2026-02-12
Home
Scan
History
Blog
My Profile


Upgrade VIP
VIP Features | VIP Features
Advanced nutritional analysis and personalized recommendations
✨ Unlimited AI Food Reports
Recipe Recommendations
Calorie Counting
Deep Analysis
Recipe Recommendations
Based on your allergies and dietary preferences
Vegetable Pasta
Vegetarian · 30 mins · Gluten-free option
VIP Exclusive
Low Salt Chicken Salad
High Protein · 15 mins · Nut-free
VIP Exclusive
VIP Membership
Monthly: HK$38 | Yearly: HK$298 (Save 30%)
Upgrade VIP


Select Plan
Monthly Plan
HK$38
Yearly Plan
HK$298
Save 30%
Select Payment Method
PayMe
Alipay HK
WeChat Pay HK
VISA / MasterCard
PayPal
Pay Now

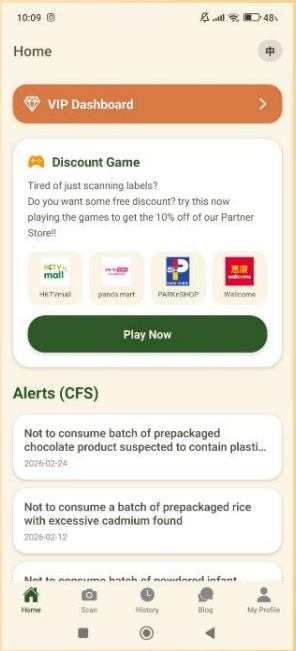

Select Plan
Monthly Plan
HK$38
Yearly Plan
HK$298
Save 30%
Select Payment Method
Payment Successful
Transaction ID: 500798836
OK
VISA / MasterCard
PayPal
Pay Now

VIP Upgrade





AND

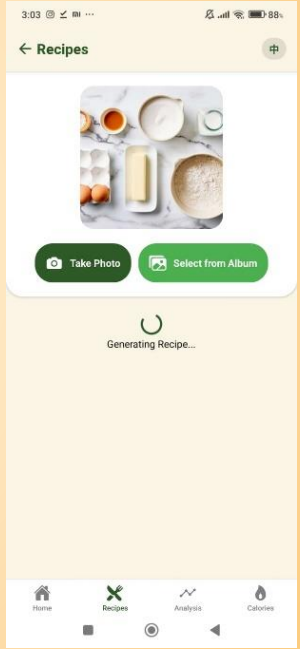


VIP Dashboard

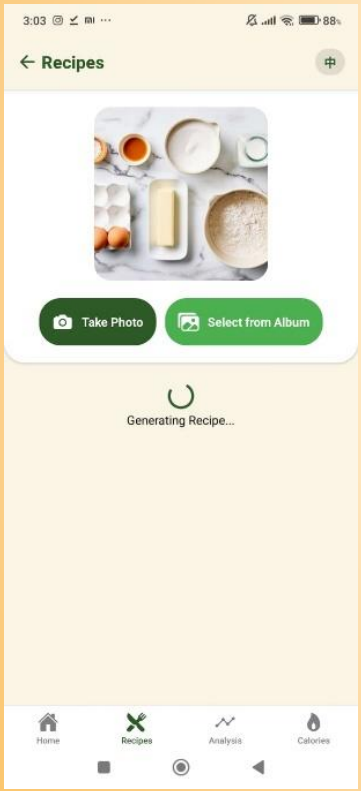


OR

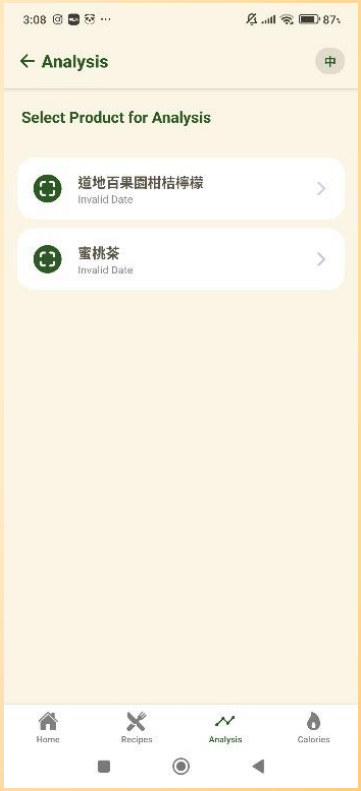




VIP Dashboard

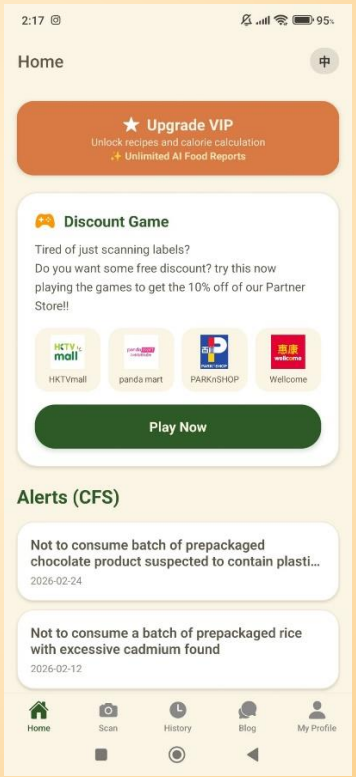






Language


2:17
95%
首頁
ENG
★ 升級 VIP
解鎖食譜推薦與卡路里計算
✨ 無限制 AI 食物報告
折扣遊戲
厭倦了只是掃描標籤？
想要一些免費折扣嗎？現在就試試
玩遊戲即可獲得我們合作夥伴商店的 10% 折扣！！
HKTVmall
panda mart
百佳
惠康
立即遊玩
食安警示 (CFS)
不要食用一批可能含有塑膠碎片的預先包裝朱古力產品
2026-02-24
不要食用一批鎘含量超出法例標準的預先包裝米
2026-02-12
首頁
掃描
歷史記錄
Blog
個人資料

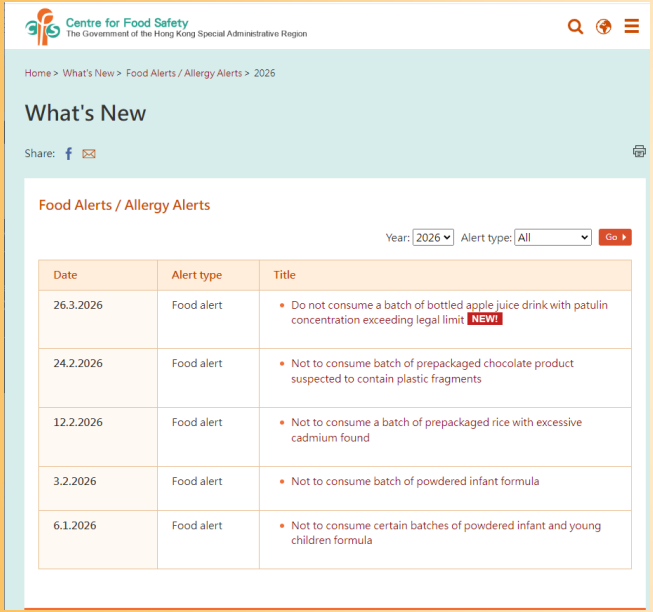

2:17
95%
Home
中
★ Upgrade VIP
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05

Demonstration



Alert news



Alert news update scheduler

```
cron.schedule('0 3 * * *', async () => {
    logger.info('Running scheduled daily update...');
    try {
        await updateAlerts();
        logger.info('Scheduled update completed successfully.');
    } catch (error) {
        logger.error(`Scheduled update failed: ${error.message}`);
    }
});
```

開放數據平台
DATA.GOV.HK

Get allergy alerts

At 0300am every days



AI prompt-Quiz game

```
const prompt = `
  Generate 5 fun, engaging, and educational multiple-choice quiz questions about food additives, food safety, and nutrition.

  Tailor the questions to a user with:
  - Allergies: ${allergies}
  - Diet: ${diet}
  - Health Status: ${healthStatus}
  - Age: ${age}

  Make sure the correct answers are scientifically accurate but explained simply.
  Focus on "hidden dangers", "healthy alternatives", or "facts about [User's Allergies/Diet]".

  Strictly conform to this JSON structure:
  [
    {
      "id": 1,
      "en": {
        "question": "English Question text here?",
        "options": ["Option A", "Option B", "Option C", "Option D"],
        "answer": "Option A"
      },
      "zh": {
        "question": "中文問題在這裏?",
        "options": ["選項 A", "選項 B", "選項 C", "選項 D"],
        "answer": "選項 A"
      }
    }
  ]

  Return ONLY valid JSON. No markdown formatting.
`;
```

- **Provide user data for analysis**
- **Request**

- **Expected output format**

AI prompt-Product analysis report

```
const prompt = `
  Analyze the following food item based on the provided text/label information.
  Item Name: ${itemName}
  Ingredients/Nutrition/Label Text: ${nutritionInformation}

  User Profile:
  Allergies: ${allergies}
  Dietary Preferences: ${diet}
  Health Status: ${healthStatus}
  Age: ${age}
  Height: ${height}
  Weight: ${weight}

  Tasks:
  1. Extract and summarize the key Food Label Information.
  2. Analyze for high salt, high sugar, or high fat content considering the user's health status (e.g., Diabetes, Hypertension).
  3. Identify any dangerous food additives.
  4. CRITICAL: Check if the item contains any ingredients matching the user's allergies, dietary preferences, or contraindicated by their health status.

  Determine "safetyLevel":
  - "Danger": Immediate harm risk (e.g. Contains Allergen).
  - "Warning": Long-term health risk or specific concern (e.g. High Sugar for Diabetic user, High Salt for Hypertension).
  - "Safe": No specific issues found.

  Provide output separated for English and Traditional Chinese (繁體中文).

  Return ONLY a JSON object with this EXACT structure:
  {
    "safetyLevel": "Danger" | "Warning" | "Safe",
    "english": {
      "labelInfo": "Extracted summary...",
      "summary": "Brief analysis...",
      "warning": "Specific warning text if any, or explanation of safety level",
      "details": "Detailed analysis..."
    },
    "chinese": {
      "labelInfo": "...",
      "summary": "...",
      "warning": "...",
      "details": "..."
    }
  }
`;
```

- **Provide user data for analysis**
- **Request**

- **Expected output format**

Reset Password (By sending OTP)



Nodemailer

Send OTP






```
const transporter = nodemailer.createTransport({
  service: 'gmail',
  auth: {
    user: 'michaelwong88041356@gmail.com',
    pass: 'paes efyf nluc fgok'
  },
  tls: {
    rejectUnauthorized: false
  }
});

app.post('/api/forgot-password', async (req, res) => {
  const { email } = req.body;
  if (!email) return res.status(400).json({ message: "Email is required" });

  try {
    const user = await User.findOne({ email });
    if (!user) return res.status(404).json({ message: "User not found" });

    const otp = Math.floor(100000 + Math.random() * 900000).toString();
    user.resetOTP = otp;
    user.resetOTPExpires = Date.now() + 10 * 60 * 1000;
    await user.save();

    const mailOptions = {
        from: 'Food Scanner App <no-reply@foodscanner.com>',
        to: email,
        subject: 'Password Reset OTP',
        text: `Your OTP for password reset is: ${otp}\nIt expires in 10 minutes.`
    };

    try {
        await transporter.sendMail(mailOptions);
    } catch (mailError) {
        console.error("Email send failed:", mailError);
    }

    res.json({ message: "OTP sent to your email" });
  } catch (error) {
    console.error("Forgot password error:", error);
    res.status(500).json({ message: "Internal Server Error" });
  }
});
```

06

Limitations



Limitations



1. OCR technique

- **Product image quality is depended by the user**

2. Lack of reliable database

- **Current approach: Scan database Queuing method(barcode)**



```
const ProductSchema = new mongoose.Schema({
  "item name": String,
  "item code": String,
  "nutrition information": String
}, { strict: false });

const Product = mongoose.model('Product', ProductSchema, 'FoodLabelBarCodeScan');

const ScanHistorySchema = new mongoose.Schema({
  "item name": String,
  "item code": String,
  "nutrition information": String,
  "user id": String,
  "warning": String,
  "health analysis": String
}, { strict: false });

const ScanHistory = mongoose.model('ScanHistory', ScanHistorySchema, 'ScanHistory');
```

```
let baseProduct = await Product.findOne({ "item code": barcode });

if (!baseProduct) {
  baseProduct = await ScanHistory.findOne({ "item code": barcode });
}
```

07

Future Improvements



Future Improvements

1. Reliable database
   - Cooperate with professionals

2. More system language

3. Attract more users to use the app
   - Cooperate with the real shop

4. OCR technique improve



08

Q&A











Thank You!

